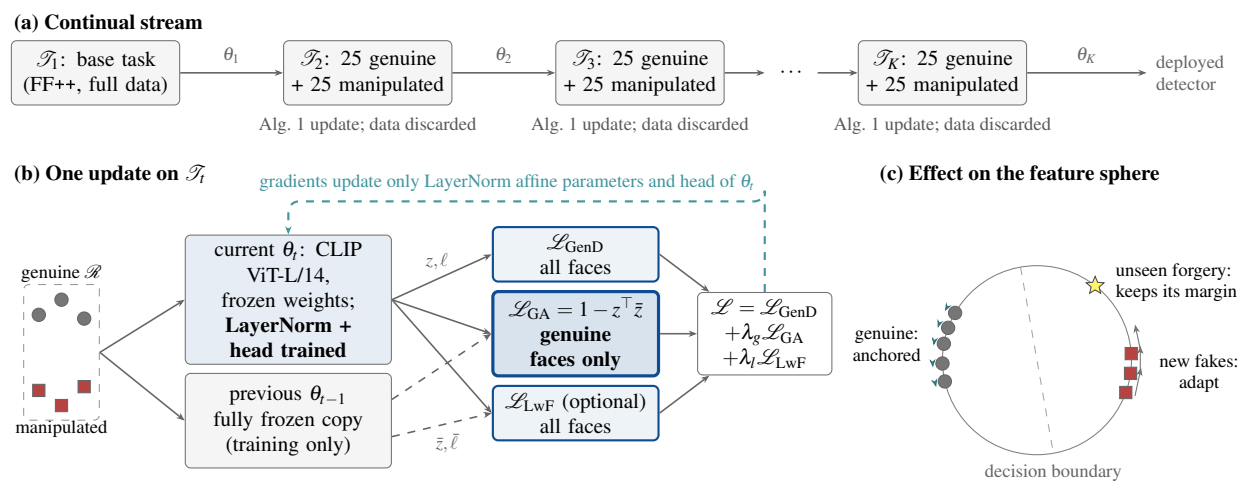

(a) Continual stream
$\mathcal{T}_1$: base task (FF++, full data)
θ_1
$\mathcal{T}_2$: 25 genuine + 25 manipulated
θ_2
$\mathcal{T}_3$: 25 genuine + 25 manipulated
...
$\mathcal{T}_K$: 25 genuine + 25 manipulated
θ_K
deployed detector
Alg. 1 update; data discarded
Alg. 1 update; data discarded
Alg. 1 update; data discarded
(b) One update on $\mathcal{T}_t$
gradients update only LayerNorm affine parameters and head of θ_t
genuine $\mathcal{R}$
manipulated
current θ_t: CLIP ViT-L/14, frozen weights; LayerNorm + head trained
previous θ_{t-1} fully frozen copy (training only)
z, ℓ
$\bar{z}, \bar{\ell}$
$\mathcal{L}_{\text{GenD}}$ all faces
$\mathcal{L}_{\text{GA}} = 1 - z^\top \bar{z}$ genuine faces only
$\mathcal{L}_{\text{LwF}}$ (optional) all faces
$\mathcal{L} = \mathcal{L}_{\text{GenD}} + \lambda_g \mathcal{L}_{\text{GA}} + \lambda_l \mathcal{L}_{\text{LwF}}$
(c) Effect on the feature sphere
unseen forgery: keeps its margin
genuine: anchored
new fakes: adapt
decision boundary